ASGHINLABS ACADEMY

Physics

FULL LESSON NOTES

COMPLETE · LESSONS 1-52

LOWER SIXTH

GCE A Level · MINESEC Progression

Every lesson in the exact order of the official MINESEC progression sheet, with diagrams and worked examples

LEARN · PRACTISE · SUCCEED

By **Mr Ayemelong S. Ghislain**

www.asghinlabs.com

Academic Year **2026-2027**



ASGHINLABS

L6

ASGHINLABS ACADEMY

Lesson notes · Workbooks · Interactive lessons · AI Tutor

AsghinLabs Academy helps Cameroonian students prepare for the GCE with study materials that follow the official MINESEC progression. These notes are one part of the collection. Everything below is available from our website.

Workbooks available

Subject	What is available
Mathematics	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
Computer Science	Form 1, Form 2 and Form 3 workbooks · Complete O Level pack · Complete A Level pack (Lower & Upper Sixth)
ICT	Complete A Level pack (ICT 0796) with GCE past questions

How to order: pick your subject and class on the website, tap "Add to order", then send the order on WhatsApp. No account is needed and no payment is taken on the website. We reply to confirm the total, payment and delivery. Schools and classes ordering 10 copies or more can ask for a group price.

Order online: www.asghinlabs.com/academy.html#workbooks

Order on WhatsApp: +237 682 402 876

ASGHINLABS AI TUTOR

Stuck on an example or a practice question in these notes? Ask the AsghinLabs AI Tutor to explain it step by step, at any time of day.

Open the AI Tutor: www.asghinlabs.com/tutor.html

More from AsghinLabs Academy

- **Full lesson notes:** Lower and Upper Sixth Mathematics, Pure Mathematics with Mechanics, Biology, Physics and Chemistry, aligned with the MINESEC progression
- **Free interactive lessons:** www.asghinlabs.com/lessons.html
- **Free notes and resources:** www.asghinlabs.com/academy.html#resources
- **Private classes and GCE exam preparation:** www.asghinlabs.com/academy.html
- **WhatsApp channel (updates and new materials):** Join the AsghinLabs channel

www.asghinlabs.com · WhatsApp +237 682 402 876 · asghinlabs@gmail.com

Table of Contents

How to use these notes 7

Physical Quantities (Lessons 1–3) 8

Lesson 1: Physical quantities, base and derived quantities, SI units 8
Physical quantities · base and derived quantities · SI units and prefixes · converting units

Lesson 2: Dimensions, physical equations and homogeneity 10
Dimensions of quantities · homogeneity of equations · deriving relationships by dimensional analysis

Lesson 3: Experimental physics: techniques and the analysis of data 11
Measuring instruments · systematic and random errors · accuracy and precision · uncertainties · graphs

More Worked Examples: Physical Quantities, Units and Experimental Physics 12
Extra worked examples and model answers for Physical Quantities, Units and Experimental Physics

Review Exercise: Physical Quantities, Units and Experimental Physics 13
Mixed exam-style questions with answers covering the whole topic

Temperature (Lessons 4–5) 14

Lesson 4: Temperature and heat; thermal equilibrium; thermometric properties 14
Temperature vs heat · thermal equilibrium · zeroth law · thermometric properties

Lesson 5: Temperature scales and types of thermometers 15
Celsius and thermodynamic scales · fixed points · liquid-in-glass, resistance, thermocouple and gas thermometers

More Worked Examples: Temperature 16
Extra worked examples and model answers for Temperature

Review Exercise: Temperature 17
Mixed exam-style questions with answers covering the whole topic

Kinematics (Lessons 6–9) 18

Lesson 6: Motion: distance, displacement, speed, velocity, acceleration, momentum and graphs 18
Distance and displacement · speed and velocity · acceleration · momentum · motion graphs

Lesson 7: Equations of uniformly accelerated motion 20
The equations of uniformly accelerated motion and their use

Lesson 8: Motion under gravity; measuring g by free fall 21
Free fall · acceleration due to gravity · measuring g with a light gate or electromagnet

Lesson 9: Projectile motion 22
Independence of horizontal and vertical motion · time of flight, greatest height and range

More Worked Examples: Kinematics 23
Extra worked examples and model answers for Kinematics

Review Exercise: Kinematics 24
Mixed exam-style questions with answers covering the whole topic

Energy Transfer: Thermal Physics (Lessons 10–11) 25

Lesson 10: Heat capacity and specific heat capacity 25
Heat capacity and specific heat capacity · electrical and mixture methods

Lesson 11: Latent heat and specific latent heat 27
Latent heat of fusion and vaporisation · measuring specific latent heat · molecular explanation

Dynamics and Force (Lessons 12–13) 28

Lesson 12: Forces: nature, types, centre of gravity, free-body diagrams and resultants 28
Fundamental forces · types of contact forces · centre of gravity · free-body diagrams · resultant forces

Lesson 13: Turning effect of forces: moments, couples and equilibrium 30
Moments · principle of moments · couples and torque · conditions for equilibrium

More Worked Examples: Forces, Moments and Equilibrium 31
Extra worked examples and model answers for Forces, Moments and Equilibrium

Review Exercise: Forces, Moments and Equilibrium 32
Mixed exam-style questions with answers covering the whole topic

Energy Transfer: Thermal Physics (Lesson 14) 33

Lesson 14: Heating and cooling curves 33
Heating and cooling curves · reading changes of state · energy calculations from curves

Newton's Laws of Motion (Lesson 15) 34

Lesson 15: Newton's first and second laws; impulse 34
Newton's first and second laws · F = ma · impulse and force–time graphs

Energy Transfer: Thermal Physics (Lessons 16–17) 35

Lesson 16: Thermal conduction: mechanism and temperature gradient 35
Mechanism of conduction · temperature gradient · good and poor conductors

Lesson 17: Thermal conductivity; comparing electrical and thermal conduction 36
Thermal conductivity equation · insulation · analogy between thermal and electrical conduction

Newton's Laws of Motion and Momentum (Lesson 18) 37

Lesson 18: Newton's third law and conservation of momentum 37
Newton's third law · action–reaction pairs · conservation of linear momentum

Energy Transfer: Thermal Physics (Lesson 19) 38

Lesson 19: Thermal convection and radiation 38
Convection currents · radiation and black bodies · Stefan's law outline · preventing heat transfer

More Worked Examples: Energy Transfer: Thermal Physics 39
Extra worked examples and model answers for Energy Transfer: Thermal Physics

Review Exercise: Energy Transfer: Thermal Physics 40
Mixed exam-style questions with answers covering the whole topic

Newton's Laws of Motion and Momentum (Lessons 20–21) 41

Lesson 20: Experiment to investigate conservation of momentum 41
Investigating momentum with trolleys, air tracks and light gates

Lesson 21: Elastic and inelastic collisions; explosions; head-on and oblique collisions 42
Elastic and inelastic collisions · explosions and recoil · oblique collisions

More Worked Examples: Newton's Laws of Motion and Momentum 43
Extra worked examples and model answers for Newton's Laws of Motion and Momentum

Review Exercise: Newton's Laws of Motion and Momentum 44
Mixed exam-style questions with answers covering the whole topic

Electrical Energy (Lessons 22–23) 45

Lesson 22: Electric current, charge, potential difference and drift velocity 45
Current as rate of flow of charge · potential difference · drift velocity I = nAvq

Lesson 23: Resistance, colour codes, Ohm's law and resistivity 46
Resistance · resistor colour code · Ohm's law · I–V characteristics · resistivity

Work, Energy and Power (Lessons 24–25) 47

Lesson 24: Work, potential energy, kinetic energy and conservation of mechanical energy 47
Work done by a force · gravitational PE and KE · conservation of mechanical energy

Lesson 25: The work–energy theorem, efficiency and power 48
Work–energy theorem · power P = W/t = Fv · efficiency

More Worked Examples: Work, Energy and Power 49
Extra worked examples and model answers for Work, Energy and Power

Review Exercise: Work, Energy and Power 50
Mixed exam-style questions with answers covering the whole topic

Electrical Energy (Lessons 26–27) 51

Lesson 26: Experiment to determine resistivity; temperature coefficient of resistance 51
Measuring resistivity with a micrometer and meters · temperature coefficient of resistance

Lesson 27: Resistor networks and the potential divider 52
Series and parallel resistors · potential dividers · thermistors and LDRs in dividers

Circular Motion (Lessons 28–29) 53

Lesson 28: Circular motion: period, frequency, angular speed, centripetal acceleration and force 53
Angular displacement, angular speed, period and frequency · centripetal acceleration and force

Lesson 29: Motion in a vertical circle, conical pendulum, cornering and banking 54
Vertical circles · conical pendulum · banking of roads and aircraft

More Worked Examples: Circular Motion 55
Extra worked examples and model answers for Circular Motion

Review Exercise: Circular Motion 56
Mixed exam-style questions with answers covering the whole topic

Electrical Energy (Lessons 30–31) 57

Lesson 30: Ideal and non-ideal ammeters and voltmeters 57
Ideal meters · effect of real meter resistance · loading effect of voltmeters

Lesson 31: E.m.f., terminal p.d. and internal resistance 58
E.m.f. and terminal p.d. · internal resistance · lost volts · measuring E and r

Simple Harmonic Motion (Lessons 32–33) 59

Lesson 32: Simple harmonic motion: meaning and equations 59
Definition of SHM · $a = -\omega^2 x$ · displacement, velocity and acceleration equations · period

Lesson 33: Energy in SHM; examples of simple harmonic oscillators 61
Kinetic and potential energy in SHM · energy graphs · mass–spring and pendulum systems

Electrical Energy (Lessons 34–35) 62

Lesson 34: Kirchhoff's laws 62
Kirchhoff's junction and loop rules · solving multi-loop circuits

Lesson 35: The potentiometer 63
Principle of the potentiometer · comparing e.m.f.s · measuring internal resistance

Simple Harmonic Motion (Lesson 36) 64

Lesson 36: Mechanical oscillations, damping and resonance 64
Free, damped and forced oscillations · resonance · useful and harmful resonance

More Worked Examples: Simple Harmonic Motion and Oscillations 65
Extra worked examples and model answers for Simple Harmonic Motion and Oscillations

Review Exercise: Simple Harmonic Motion and Oscillations 66
Mixed exam-style questions with answers covering the whole topic

Electrical Energy (Lesson 37) 67

Lesson 37: The Wheatstone bridge 67
Wheatstone bridge balance condition · the metre bridge · applications

More Worked Examples: Electrical Energy: Current Electricity 68
Extra worked examples and model answers for Electrical Energy: Current Electricity

Review Exercise: Electrical Energy: Current Electricity 69
Mixed exam-style questions with answers covering the whole topic

Solids and Liquids (Lessons 38–39) 70

Lesson 38: Molecular properties of solids, liquids and gases; force and potential energy curves 70
Arrangement and motion of particles in solids, liquids and gases · force–separation and energy–separation curves

Lesson 39: Elasticity, Young modulus and energy stored in a stretched wire 72
Hooke's law · stress, strain, Young modulus · stress–strain curves · strain energy

Energy Resources and Environmental Physics (Lesson 40) 74

Lesson 40: Classification of energy sources; efficiency of conversion 74
Renewable and non-renewable sources · energy conversions · efficiency of energy chains

Solids and Liquids (Lesson 41) 75

Lesson 41: Experiment to determine the Young modulus 75
Searle's apparatus · measurements and graph · precautions

Energy Resources and Environmental Physics (Lesson 42) 76

Lesson 42: Hydroelectricity and wind energy 76
Hydroelectric power $P = \rho Qgh$ · wind power $\frac{1}{2}\rho Av^3$ · advantages and disadvantages

Solids and Liquids (Lesson 43) 77

Lesson 43: Surface tension and capillarity; angle of contact 77
Surface tension · angle of contact · capillary rise and depression

More Worked Examples: Solids and Liquids 79
Extra worked examples and model answers for Solids and Liquids

Review Exercise: Solids and Liquids 80
Mixed exam-style questions with answers covering the whole topic

Energy Resources and Environmental Physics (Lessons 44–45) 81

Lesson 44: Solar energy and tidal energy 81
Solar cells and solar heating · tidal power · advantages and disadvantages

Lesson 45: Biomass, geothermal energy and wave energy 82
Biomass and biogas · geothermal energy · wave energy

Gases and Thermodynamics (Lessons 46–47) 83

Lesson 46: Brownian motion, gas laws and the ideal gas equation 83
Brownian motion · Boyle's, Charles's and pressure laws · ideal gas equation

Lesson 47: Kinetic theory of ideal gases; distribution of speeds; real gases 84
Assumptions of kinetic theory · $pV = \frac{1}{3}Nm\langle c^2 \rangle$ · mean KE and temperature · r.m.s. speed · real gases

Energy Resources and Environmental Physics (Lessons 48–49) 86

Lesson 48: Fossil fuels and nuclear fuel 86
Formation and use of fossil fuels · nuclear fission fuel · $E = mc^2$ · pollution and waste

Lesson 49: Structure of the atmosphere; ozonosphere and ionosphere 87
Layers of the atmosphere · ozone layer and UV absorption · ionosphere and radio waves

Gases and Thermodynamics (Lessons 50–51) 88

Lesson 50: First law of thermodynamics; thermodynamic processes 88
First law $\Delta U = Q + W$ · work done by a gas · isothermal, adiabatic, isobaric and isovolumetric processes

Lesson 51: Second law of thermodynamics, heat engines and entropy 90
Second law · heat engines and efficiency · Carnot efficiency · refrigerators and heat pumps · entropy

More Worked Examples: Gases and Thermodynamics 91
Extra worked examples and model answers for Gases and Thermodynamics

Review Exercise: Gases and Thermodynamics 92
Mixed exam-style questions with answers covering the whole topic

Energy Resources and Environmental Physics (Lesson 52) 93

Lesson 52: The greenhouse effect, global warming and climate change 93
Greenhouse gases · enhanced greenhouse effect · global warming · effects and mitigation

More Worked Examples: Energy Resources and Environmental Physics 94
Extra worked examples and model answers for Energy Resources and Environmental Physics

Review Exercise: Energy Resources and Environmental Physics 95
Mixed exam-style questions with answers covering the whole topic

Formula Summary (Lower Sixth Physics) 96

How to use these notes

These notes follow the official **MINESEC progression sheet** lesson by lesson. Each lesson has its own section, numbered exactly as in the progression, so you always know where you are in the syllabus.

Feature	What to do with it
Learning objectives (green box)	Read them first: they tell you what you must be able to do by the end of the lesson.
Lesson notes and Key Points (blue box)	Learn the definitions and formulas in the Key Points by heart.
Exam tips (yellow box)	Common mistakes and how GCE examiners expect answers to be set out.
Worked examples	Cover the solution, try the question yourself, then compare line by line.
Practice questions	Do them without looking at the answers. Mark your work and redo any you got wrong.
Review exercise	Mixed exam-style questions to test yourself on several lessons at once.

Need help? Ask the AsghinLabs AI Tutor at **www.asghinlabs.com/tutor.html** or message us on WhatsApp (+237 682 402 876).

Physical Quantities (Lessons 1–3)

In these notes take g = 9.81 m s^{-2} unless a question gives another value.

Lesson	Title (MINESEC progression)
1	Physical quantities, base and derived quantities, SI units
2	Dimensions, physical equations and homogeneity
3	Experimental physics: techniques and the analysis of data

Lesson 1: Physical quantities, base and derived quantities, SI units

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish between base and derived quantities and name the seven SI base units
- ✔ express derived units in base units
- ✔ use prefixes and convert between units

Lesson notes

A **physical quantity** is a property that can be measured and expressed as a number multiplied by a unit, e.g. length = 2.5 m.

KEY POINTS

The seven SI base quantities and units:

Base quantity	Symbol	SI base unit	Unit symbol
Mass	m	kilogram	kg
Length	l	metre	m
Time	t	second	s
Electric current	I	ampere	A
Thermodynamic temperature	T	kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	I_v	candela	cd

Derived quantities are defined from base quantities; their units are combinations of base units.

Derived quantity	Definition	Unit	In base units
Velocity	displacement ÷ time	m s^{-1}	m s^{-1}
Acceleration	change in velocity ÷ time	m s^{-2}	m s^{-2}
Force	mass × acceleration	newton (N)	kg m s^{-2}
Energy, work	force × distance	joule (J)	kg m^2 s^{-2}
Power	work ÷ time	watt (W)	kg m^2 s^{-3}
Pressure	force ÷ area	pascal (Pa)	kg m^{-1} s^{-2}
Charge	current × time	coulomb (C)	A s
Potential difference	energy ÷ charge	volt (V)	kg m^2 s^{-3} A^{-1}

Derived quantity	Definition	Unit	In base units
Resistance	p.d. ÷ current	ohm (Ω)	kg m^2 s^{-3} A^{-2}
Frequency	1 ÷ period	hertz (Hz)	s^{-1}

KEY POINTS

Prefixes: pico p 10^{-12}; nano n 10^{-9}; micro μ 10^{-6}; milli m 10^{-3}; centi c 10^{-2}; kilo k 10^{3}; mega M 10^{6}; giga G 10^{9}; tera T 10^{12}

- **Scalars** have magnitude only (mass, time, energy, speed); **vectors** have magnitude and direction (displacement, velocity, force, momentum).

Worked examples

Example: Express the joule in SI base units.

J = N × m = (kg m s^{-2}) × m = kg m^2 s^{-2}

Example: Convert 72 km h^{-1} to m s^{-1} and 450 nm to m.

72 × 1000 ÷ 3600 = 20 m s^{-1}; 450 nm = 450 × 10^{-9} = 4.5 × 10^{-7} m

Practice Questions: Lesson 1

1. Express the newton and the watt in SI base units.
2. Convert (a) 54 km h^{-1} to m s^{-1} (b) 2.5 mm^2 to m^2 (c) 3.0 g cm^{-3} to kg m^{-3}.

Answers: *1. N = kg m s^{-2}; W = kg m^2 s^{-3} 2. (a) 15 m s^{-1} (b) 2.5 × 10^{-6} m^2 (c) 3.0 × 10^{3} kg m^{-3}*

Lesson 2: Dimensions, physical equations and homogeneity

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ write the dimensions of physical quantities
- ✔ check the homogeneity of equations
- ✔ use dimensions to predict the form of an equation

Lesson notes

The **dimensions** of a quantity show how it depends on the base quantities mass [M], length [L], time [T] (and current [I], temperature [θ]).

KEY POINTS

[velocity] = LT^{-1}; [acceleration] = LT^{-2}; [force] = MLT^{-2}; [energy] = ML^2T^{-2}; [power] = ML^2T^{-3}; [pressure] = $ML^{-1}T^{-2}$; [density] = ML^{-3}

An equation is **homogeneous** (dimensionally consistent) if every term has the same dimensions. A correct equation must be homogeneous, but a homogeneous equation may still be wrong (e.g. a wrong numerical constant).

Uses of dimensions: checking equations; finding units; deriving the form of a relationship (dimensional analysis).

Worked examples

Example: Show that $v^2 = u^2 + 2as$ is homogeneous.

$[v^2] = [u^2] = L^2T^{-2}$; $[2as] = (LT^{-2})(L) = L^2T^{-2}$ (numbers have no dimensions). All terms match.

Example: The period T of a simple pendulum may depend on its mass m, length l and g. Use dimensions to find the form of the equation.

Let $T = k\,m^x\,l^y\,g^z$: $T^1 = M^x\,L^y\,(LT^{-2})^z$

M: $x = 0$; T: $1 = -2z$, $z = -\tfrac{1}{2}$; L: $0 = y + z$, $y = \tfrac{1}{2}$

$T = k\sqrt{(l/g)}$ (experiment gives $k = 2\pi$; the period does not depend on mass)

Practice Questions: Lesson 2

1. Show that $s = ut + \tfrac{1}{2}at^2$ is dimensionally consistent.
2. The speed v of a wave on a string depends on the tension T and the mass per unit length μ. Use dimensions to show that $v = k\sqrt{(T/\mu)}$.

Answers: *1. every term has dimension L 2. [T] = MLT^{-2}, [μ] = ML^{-1}; T/μ has dimensions L^2T^{-2}, so √(T/μ) has dimensions LT^{-1}*

Lesson 3: Experimental physics: techniques and the analysis of data

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish between systematic and random errors, accuracy and precision
- ✔ calculate absolute and percentage uncertainties and combine them
- ✔ analyse data using straight-line graphs

Lesson notes

Structure of an experiment: aim → apparatus and set-up (diagram) → procedure → observations and readings (table) → analysis (graph, calculation) → conclusion → precautions and sources of error.

- **Measuring instruments:** metre rule (±1 mm), vernier calipers (±0.1 mm or 0.02 mm), micrometer screw gauge (±0.01 mm), stopwatch, balance, thermometer, ammeter, voltmeter.
- **Errors: random errors** (scatter of readings, reduced by repeating and averaging; e.g. reaction time) and **systematic errors** (all readings shifted in the same direction, not reduced by repeating; e.g. zero error, parallax always from the same side, wrong calibration).
- **Accuracy:** closeness to the true value. **Precision:** closeness of repeated readings to each other.

KEY POINTS

Percentage uncertainty = (absolute uncertainty ÷ measured value) × 100%. When quantities are multiplied or divided, ADD the percentage uncertainties; for a power, multiply the percentage uncertainty by the power.

Tables and graphs: column headings with quantity and unit (e.g. t / s); consistent significant figures; graphs with labelled axes, sensible scales (plot occupies more than half the grid), best-fit line; gradient from a large triangle.

Precautions: avoid parallax (eye level perpendicular to the scale); check zero errors; repeat readings; keep other variables constant; switch off the current between readings to avoid heating.

Worked examples

Example: The diameter of a wire is 0.50 ± 0.01 mm. Calculate the percentage uncertainty in its cross-sectional area.

% uncertainty in d = 0.01/0.50 × 100 = 2%

$A = \pi d^2/4$, so % uncertainty in A = 2 × 2% = 4%

Example: The time for 20 oscillations is 32.4 s, measured with a reaction-time uncertainty of ±0.2 s. Find the period and its uncertainty.

T = 32.4/20 = 1.62 s; uncertainty = 0.2/20 = 0.01 s, so T = 1.62 ± 0.01 s

(Timing many oscillations reduces the uncertainty in each.)

Practice Questions: Lesson 3

1. A length is 25.0 ± 0.1 cm and a width is 10.0 ± 0.1 cm. Find the area and its percentage uncertainty.
2. Distinguish between accuracy and precision.

Answers: *1. 250 cm^2 ± 1.4% (± 3.5 cm^2) 2. accuracy: closeness to the true value; precision: closeness of repeated readings to each other (small spread)*

More Worked Examples: Physical Quantities, Units and Experimental Physics

Example: Show that the equation for pressure in a liquid, $p = \rho gh$, is homogeneous.

$[p] = \text{force/area} = MLT^{-2}/L^2 = ML^{-1}T^{-2}$

$[\rho gh] = (ML^{-3})(LT^{-2})(L) = ML^{-1}T^{-2}$

Both sides have the same dimensions, so the equation is homogeneous.

Example: Convert 72 km h^{-1} to m s^{-1} and 2.5 g cm^{-3} to kg m^{-3}.

72 km h^{-1} = 72 × 1000 m/3600 s = 20 m s^{-1}

2.5 g cm^{-3} = 2.5 × 10^{-3} kg/10^{-6} m^3 = 2500 kg m^{-3}

Example: A metal cube has side 2.00 ± 0.01 cm and mass 50.0 ± 0.1 g. Find its density and the uncertainty.

Density = 50.0/2.00^3 = 6.25 g cm^{-3}

% uncertainty in side = 0.5%; volume depends on side^3, so 3 × 0.5% = 1.5%

% uncertainty in mass = 0.1/50.0 × 100 = 0.2%

Total = 1.5% + 0.2% = 1.7%; absolute = 0.017 × 6.25 = 0.11

Density = 6.25 ± 0.11 g cm^{-3}

Example: The speed of sound v in a gas depends on the pressure p and density ρ. Use dimensions to find the form of the equation.

Let $v = k\,p^x\,\rho^y$: $LT^{-1} = (ML^{-1}T^{-2})^x\,(ML^{-3})^y$

M: 0 = x + y. T: −1 = −2x, so x = ½ and y = −½. L: −x − 3y = −½ + 3/2 = 1 ✓

$v = k\sqrt{(p/\rho)}$

Example: A micrometer screw gauge shows 5.5 mm on the main (sleeve) scale and the 28th division of the thimble against the line. Each thimble division is 0.01 mm. Give the reading.

Reading = 5.5 + 28 × 0.01 = 5.78 mm (check for zero error first)

Review Exercise: Physical Quantities, Units and Experimental Physics

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. Express the pascal and the watt in SI base units.
2. Check the homogeneity of $s = ut + \tfrac{1}{2}at^2$.
3. The speed v of waves on a string depends on the tension F and the mass per unit length μ. Use dimensions to show that $v = k\sqrt{(F/\mu)}$.
4. A cube has sides measured as 2.0 ± 0.1 cm. Find the percentage uncertainty in its volume.
5. Distinguish between random and systematic errors with one example of each.

Answers: *1. Pa = kg m^{-1} s^{-2}, W = kg m^2 s^{-3} 2. each term has dimension L 3. [F] = MLT^{-2}, [μ] = ML^{-1}; [F/μ] = L^{2}T^{-2} 4. 15% 5. see Lesson 3*

Temperature (Lessons 4–5)

Lesson	Title (MINESEC progression)
4	Temperature and heat; thermal equilibrium; thermometric properties
5	Temperature scales and types of thermometers

Lesson 4: Temperature and heat; thermal equilibrium; thermometric properties

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish between temperature and heat
- ✔ state the zeroth law of thermodynamics
- ✔ describe thermometric properties

Lesson notes

- **Heat** is energy transferred from a hotter body to a colder one because of a temperature difference (unit: joule).
- **Temperature** measures how hot a body is; it determines the direction of heat flow. On the molecular scale, it is related to the average kinetic energy of the particles.
- **Thermal equilibrium:** two bodies in contact with no net heat flow between them are at the same temperature.

KEY POINTS

Zeroth law of thermodynamics: if two bodies are each in thermal equilibrium with a third body, they are in thermal equilibrium with each other. (This is why thermometers work.)

Thermometric property: a physical property that changes continuously and measurably with temperature: length of a liquid column, pressure of a gas at constant volume, resistance of a metal wire, e.m.f. of a thermocouple, colour of radiation from a hot body.

Good thermometric property: varies uniformly (linearly) with temperature, over a wide range, sensitively, and is easy to measure.

Practice Questions: Lesson 4

1. State the zeroth law of thermodynamics.
2. Give three thermometric properties.

Answers: *1. if A and B are each in thermal equilibrium with C, then A and B are in thermal equilibrium with each other 2. volume of a liquid, resistance of a wire, e.m.f. of a thermocouple, pressure of a gas at constant volume*

Lesson 5: Temperature scales and types of thermometers

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ calculate temperatures on an empirical scale
- ✔ convert between Celsius and kelvin
- ✔ compare different types of thermometer

Lesson notes

KEY POINTS

Empirical (centigrade) scale using fixed points (ice point 0 °C, steam point 100 °C): $\theta = (X_\theta - X_0)/(X_{100} - X_0) \times 100$ °C, where X is the thermometric property

Thermodynamic (Kelvin) scale: absolute zero = 0 K; triple point of water = 273.16 K. $T/K = \theta/°C + 273.15$

Different thermometers agree at the fixed points but may disagree in between, because their thermometric properties do not vary in the same way with temperature.

Thermometer	Property	Range	Advantages / uses
Liquid-in-glass (mercury, alcohol)	expansion of liquid	−39 to 350 °C (mercury)	simple, cheap, direct reading
Constant-volume gas thermometer	pressure of a fixed mass of gas	very wide	standard thermometer, accurate; bulky, slow
Platinum resistance	resistance of a metal	−200 to 1200 °C	accurate; slow to respond
Thermocouple	e.m.f. between two junctions of different metals	−200 to 1500 °C	small, fast response, remote reading
Thermistor	resistance of a semiconductor (falls as T rises)	−50 to 300 °C	very sensitive, small
Pyrometer	radiation emitted	above 1000 °C	no contact (furnaces)

Worked examples

Example: A resistance thermometer reads 40.0 Ω at 0 °C and 55.0 Ω at 100 °C. Find the temperature on its scale when the resistance is 47.5 Ω.

$\theta = (47.5 - 40.0)/(55.0 - 40.0) \times 100 = 50$ °C

Example: Convert 27 °C and −196 °C (liquid nitrogen) to kelvin.

$27 + 273 = 300$ K; $-196 + 273 = 77$ K

Practice Questions: Lesson 5

1. A thermocouple gives 0 mV at 0 °C and 4.0 mV at 100 °C. Find the temperature on its scale when the e.m.f. is 1.4 mV.
2. Convert 37 °C to kelvin and 350 K to °C.

Answers: *1. 35 °C 2. 310 K; 77 °C*

More Worked Examples: Temperature

Example: A constant-volume gas thermometer reads 100.0 kPa at the ice point and 136.6 kPa at the steam point. Find the temperature when it reads 120.0 kPa.

$\theta = (120.0 - 100.0)/(136.6 - 100.0) \times 100 = 20.0/36.6 \times 100 = 54.6\ ^\circ\text{C}$

Example: The length of a mercury column is 3.0 cm at 0 °C and 23.0 cm at 100 °C. What is the length at 35 °C?

Length change per °C = 20.0/100 = 0.20 cm

$l = 3.0 + 35 \times 0.20 = 10.0$ cm

Example: A thermocouple has e.m.f. $E = 40\theta - 0.040\theta^2$ μV (θ in °C on the gas scale). What temperature does the thermocouple scale give when the gas scale reads 50 °C?

$E_0 = 0$; $E_{100} = 4000 - 400 = 3600$ μV; $E_{50} = 2000 - 100 = 1900$ μV

θ(thermocouple) = 1900/3600 × 100 = 52.8 °C

The scales agree at 0 and 100 °C but not in between, because E is not proportional to θ.

Review Exercise: Temperature

Answer these mixed questions after finishing the topic. They combine ideas from several lessons, like GCE questions do.

Review Questions

1. State the zeroth law of thermodynamics and explain its importance for thermometry.
2. The length of a mercury column is 3.0 cm at 0 °C and 23.0 cm at 100 °C. Find the temperature when it is 12.0 cm.
3. Give two advantages of a thermocouple over a mercury thermometer.

Answers: *2. 45 °C 3. small heat capacity, fast response; wide range; can be read remotely (electrical output)*

Kinematics (Lessons 6–9)

Lesson	Title (MINESEC progression)
6	Motion: distance, displacement, speed, velocity, acceleration, momentum and graphs
7	Equations of uniformly accelerated motion
8	Motion under gravity; measuring g by free fall
9	Projectile motion

Lesson 6: Motion: distance, displacement, speed, velocity, acceleration, momentum and graphs

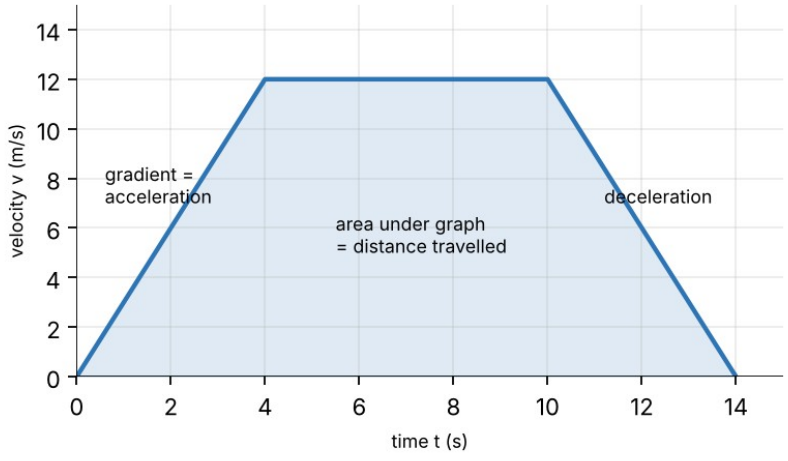
LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ distinguish between scalar and vector quantities of motion
- ✔ interpret displacement–time and velocity–time graphs

Lesson notes

Quantity	Definition	Type	Unit
Distance	total length of path	scalar	m
Displacement	distance in a stated direction from a reference point	vector	m
Speed	distance ÷ time	scalar	$m\ s^{-1}$
Velocity	rate of change of displacement	vector	$m\ s^{-1}$
Acceleration	rate of change of velocity	vector	$m\ s^{-2}$
Linear momentum	mass × velocity	vector	$kg\ m\ s^{-1}$



KEY POINTS

Gradient of a displacement–time graph = velocity. Gradient of a velocity–time graph = acceleration. Area under a velocity–time graph = displacement.

Worked examples

Example: A car travels 3 km north in 4 minutes, then 4 km east in 6 minutes. Find its average speed and average velocity.

Distance = 7 km in 10 min: average speed = 7000/600 = 11.7 m s^{-1}

Displacement = $\sqrt{(3^2 + 4^2)}$ = 5 km, at $\tan^{-1}(4/3)$ = 53° east of north: average velocity = 5000/600 = 8.3 m s^{-1} on a bearing of 053°

Practice Questions: Lesson 6

1. A cyclist travels 600 m east in 60 s and then 200 m west in 40 s. Find the average speed and average velocity.
2. What does the area under a velocity–time graph represent?

Answers: *1. 8.0 m s^{-1}; 4.0 m s^{-1} east 2. the displacement*

Lesson 7: Equations of uniformly accelerated motion

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ derive and use the equations of uniformly accelerated motion

Lesson notes

KEY POINTS

$v = u + at$; $s = ut + \tfrac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s = \tfrac{1}{2}(u + v)t$ (constant acceleration in a straight line)

Derivation: the v–t graph is a straight line of gradient a (so $v = u + at$); the area under it (a trapezium) gives $s = \tfrac{1}{2}(u + v)t$; combining gives the others.

Worked examples

Example: A car moving at 12 m s^{-1} accelerates uniformly at 1.5 m s^{-2} for 8 s. Find its final speed and the distance travelled.

$v = 12 + 1.5 \times 8 = 24$ m s^{-1}

$s = 12 \times 8 + \tfrac{1}{2} \times 1.5 \times 64 = 96 + 48 = 144$ m

Example: A motorcyclist braking from 25 m s^{-1} stops in 50 m. Find the deceleration and the time taken.

$0 = 625 + 2a(50)$, $a = -6.25$ m s^{-2} (deceleration 6.25 m s^{-2})

$t = (0 - 25)/(-6.25) = 4.0$ s

Practice Questions: Lesson 7

1. A car accelerates from rest at 2.5 m s^{-2} for 6.0 s. Find its speed and distance.
2. A train slows from 30 m s^{-1} to 10 m s^{-1} over 400 m. Find its deceleration.

Answers: *1. 15 m s^{-1}; 45 m 2. 1.0 m s^{-2}*

Lesson 8: Motion under gravity; measuring g by free fall

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ apply the equations of motion to free fall
- ✔ describe an experiment to measure g

Lesson notes

Near the Earth's surface, ignoring air resistance, all objects fall with the same acceleration **g ≈ 9.81 m s^{-2}** downwards.

Experiment to measure g: a steel ball is held by an electromagnet at height h above a trapdoor; switching off the magnet starts an electronic timer; the ball opening the trapdoor stops it. Repeat for several heights. Since $h = \tfrac{1}{2}gt^2$, a graph of h against t^2 is a straight line through the origin with gradient g/2, so g = 2 × gradient. (Using several heights and a graph removes the effect of a constant delay.)

- With air resistance, a falling body reaches **terminal velocity** when the drag force equals its weight (no resultant force).

Worked examples

Example: A stone is dropped from a bridge and hits the water 2.5 s later. Find the height of the bridge and the speed of impact.

$h = \tfrac{1}{2} \times 9.81 \times 2.5^2 = 30.7$ m; $v = 9.81 \times 2.5 = 24.5$ m s^{-1}

Practice Questions: Lesson 8

1. A ball is dropped from 20 m. Find the time to fall and the speed on impact (g = 9.8 m s^{-2}).
2. Describe how to measure g using an electromagnet and a timer.

Answers: *1. 2.02 s; 19.8 m s^{-1} 2. release a steel ball from the electromagnet (starting the timer); it breaks a trapdoor switch (stopping the timer); repeat for several heights h; plot h against t^2: gradient = g/2*

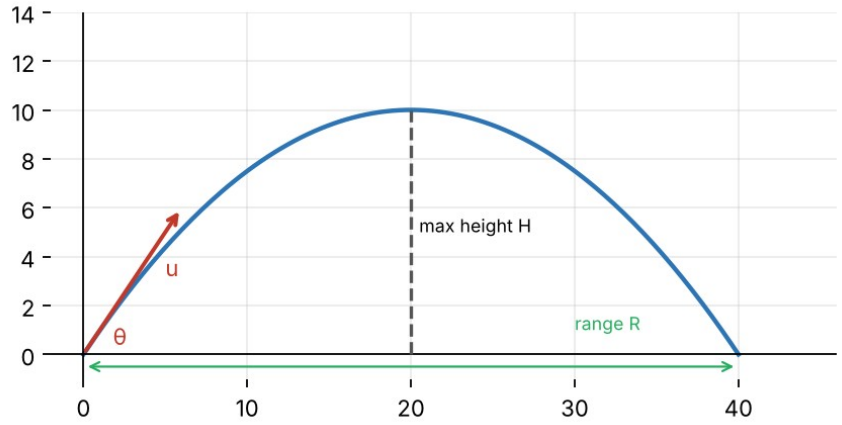
Lesson 9: Projectile motion

LEARNING OBJECTIVES

By the end of this lesson you should be able to:

- ✔ analyse projectile motion

Lesson notes



KEY POINTS

The horizontal velocity u cos θ is constant (no horizontal force); the vertical motion has acceleration g downwards. The two motions are independent.

Time of flight $T = 2u \sin\theta/g$; maximum height $H = u^2 \sin^2\theta/(2g)$; range $R = u^2 \sin 2\theta/g$ (on level ground); maximum range at $\theta = 45°$

Worked examples

Example: A ball is kicked at 20 m s^{-1} at 35° above the horizontal. Find the time of flight, the maximum height and the range.

$u \sin\theta = 11.47$ m s^{-1}; $u \cos\theta = 16.38$ m s^{-1}

$T = 2 \times 11.47/9.81 = 2.34$ s

$H = 11.47^2/(2 \times 9.81) = 6.71$ m

$R = 16.38 \times 2.34 = 38.3$ m

Example: A ball rolls horizontally off a table 0.80 m high at 2.5 m s^{-1}. How far from the table does it land?

Vertical: $0.80 = \tfrac{1}{2}(9.81)t^2$, $t = 0.404$ s; horizontal: $x = 2.5 \times 0.404 = 1.01$ m

Practice Questions: Lesson 9

1. A ball is thrown at 15 m s^{-1} at 40° above the horizontal. Find the time of flight and the range (g = 9.8).
2. A stone is thrown horizontally at 8.0 m s^{-1} from a cliff 45 m high. Find the horizontal distance travelled.

Answers: *1. 1.97 s; 22.6 m 2. 24.2 m*

END OF FREE SAMPLE

Get the full notes

You have just read the opening of **Lower Sixth Physics**. The complete book has every lesson of the MINESEC progression sheet, in order:

- Every lesson, each with objectives, clear notes and KEY POINTS
- Worked examples and practice questions with answers
- Review exercises at the end of every chapter

This book: **3 000 FCFA** · Lower + Upper Sixth: **5 000 FCFA** per subject
Complete 5-subject A Level package: **20 000 FCFA**

Order at asghinlabs.com/academy

or WhatsApp **+237 682 402 876** · PDF sent straight to your phone